

# Fundamentals of Information Systems Security

## Chapter 1 Quiz Answer Key

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### Section A: Multiple Choice (40 points)

1. **A** — Confidentiality, Integrity, Availability
2. **D** — Business Domain (not one of the seven IT domains)
3. **B** — To reduce risk and guide security decisions
4. **C** — Social engineering attack
5. **C** — 60-80%
6. **C** — Biometric scanner
7. **B** — Authentication
8. **B** — Policy → Standards → Procedures → Guidelines
9. **C** — SYN Flood attack
10. **A** — Investigation, Analysis, Design, Implementation, Maintenance

### Section B: True/False (20 points)

1. **True** — The human factor is indeed the weakest link
2. **True** — A law is a legislative piece voted on by the people
3. **False** — LAN-to-WAN Domain deals with gateways to external networks
4. **False** — Structured attacks are performed by skilled, motivated hackers
5. **True** — Three communities of interest are involved
6. **False** — Physical sabotage is a physical threat, not digital
7. **True** — AUP defines proper resource usage
8. **False** — Internal attacks account for 60-80%
9. **True** — Shoulder surfing is a physical security threat
10. **True** — The six P's framework includes all listed elements

## Section C: Short Answer (40 points)

### 1. Structured vs. Unstructured Attacks (10 points)

- **Structured Attack:** Perpetrated by skilled, motivated hackers with custom exploit development, systematic targeting, and significant financial impact. *Example: Advanced Persistent Threat (APT) campaigns, nation-state attacks*
- **Unstructured Attack:** Conducted by inexperienced attackers (“script kiddies”) using pre-written tools and scripts, opportunistic targeting. *Example: Using pre-packaged exploits from the internet*

### 2. Three Communities of Interest (10 points)

- **Information Security Professionals:** Security managers and technical experts focused on protection mechanisms
- **Information Technology Professionals:** System administrators and network engineers focused on system functionality
- **Business Management:** Non-technical business leaders focused on business objectives and ROI

### 3. Five-Step Problem-Solving Methodology (10 points)

- (a) **Recognize and Define the Problem:** Identify security incidents and define scope
- (b) **Gather Facts and Make Assumptions:** Collect logs, evidence, and relevant data
- (c) **Develop Possible Solutions:** Brainstorm multiple approaches
- (d) **Analyze and Compare Solutions:** Evaluate effectiveness and cost
- (e) **Select, Implement, and Evaluate:** Choose optimal solution and monitor results

### 4. Four Levels of Data Classification (10 points)

- **Private Data:** Personal information requiring compliance — High security controls
- **Confidential:** Intellectual property, trade secrets — Strict access limitations
- **Internal Use Only:** Internal organizational information — Moderate controls
- **Public Domain:** Publicly shared information — Minimal controls